

NLA Note

1 Background

Floating Point ($\mathbb{F}$): A number = mantissa $\times$ base^{exponent} (e.g. $2048 = 1 \times 2^{11}$ mantissa=1, base=2, exponent=11)

Assumptions: There is $\varepsilon_m > 0$ (machine epsilon) **A1:** $\forall \alpha \in \mathbb{R}, \exists \varepsilon \in (-\varepsilon_m, \varepsilon_m)$ s.t. $fl : \mathbb{R} \rightarrow \mathbb{F}$ by $fl(\alpha) = \alpha(1 + \varepsilon)$

A2: $\forall \alpha, \beta \in \mathbb{F}, \forall * \in \{+, -, \times, \div\}, \exists \varepsilon \in (-\varepsilon_m, \varepsilon_m)$ s.t. $\alpha \circledast \beta = (\alpha * \beta)(1 + \varepsilon)$

A1: The rounding error in α is relative to the size of α ($|\alpha - fl(\alpha)| < |\alpha|\varepsilon_m$) A2: The error in $\alpha \circledast \beta$ is relative to the size of $\alpha * \beta$ ($|\alpha \circledast \beta - \alpha * \beta| \leq |\alpha * \beta|\varepsilon_m$)

IEEE 754: 64-bit: denote $\mathbb{F}$. 1-bit: sign 52-bit: mantissa 11-bit: exponent (base 2). $\varepsilon_m \approx 2.22 \times 10^{-16}$

Cost of Algorithm (C): $C :=$ number of floating point operations (FLOPs) If $C(n) \leq \alpha f(n), \Rightarrow C(n) = \mathcal{O}(f(n))$. n is problem size

Norms: $\|\mathbf{x}\|_p := (\sum_{i=1}^n |x_i|^p)^{1/p}, p \geq 1$ $\|\mathbf{x}\|_\infty := \max_{1 \leq i \leq n} |x_i|$ $\|\mathbf{x}\|_2 := (\sum_{i=1}^n |x_i|^2)^{1/2}$ $\|\mathbf{x}\|_1 := \sum_{i=1}^n |x_i|$

- $\|\mathbf{x}\|_p \geq 0$ $\|\mathbf{x}\|_p = 0 \Leftrightarrow \mathbf{x} = \mathbf{0}$
- $\|\alpha \mathbf{x}\|_p = |\alpha| \|\mathbf{x}\|_p$ for $\alpha \in \mathbb{R}$
- $\|\mathbf{x} + \mathbf{y}\|_p \leq \|\mathbf{x}\|_p + \|\mathbf{y}\|_p$ (Triangle inequality)
- Induced Norm:** $\|\mathbf{A}\|_p := \max_{\mathbf{x} \neq \mathbf{0}} \frac{\|\mathbf{Ax}\|_p}{\|\mathbf{x}\|_p}$
- $\|\mathbf{A}_{m \times n}\|_\infty := \max_{1 \leq j \leq m} \sum_{i=1}^n |a_{ij}|$ (maximum row sum)
- $\|\mathbf{A}_{m \times n}\|_1 := \max_{1 \leq i \leq n} \sum_{j=1}^m |a_{ij}|$ (maximum column sum)
- $\|\mathbf{A}_{m \times n}\|_2 := \sqrt{\rho(\mathbf{A}^\top \mathbf{A})}$ $\rho(\mathbf{A}^\top \mathbf{A}) = \max\{|\lambda| : \lambda \text{ is eigenvalue of } \mathbf{A}^\top \mathbf{A}\}$

2 Algorithms

DP Dot product $C = \sum_{i=1}^n 2 = 2n = \mathcal{O}(n)$

Input: $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$

Output: $s = \mathbf{x} \cdot \mathbf{y} = \mathbf{x}^\top \mathbf{y}$

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1:  $s = 0$ 
2: for  $i = 1, \dots, n$  do
3:    $s = s + x_i y_i$ 
4: end for
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MV Matrix-vector multiplication

$C = \sum_{i=1}^m 2n = 2nm = \mathcal{O}(mn)$

Input: $\mathbf{A} \in \mathbb{R}^{m \times n}, \mathbf{x} \in \mathbb{R}^n$

Output: $\mathbf{y} = \mathbf{Ax}$

```
1: for  $i = 1, \dots, m$  do
2:    $y_i = \mathbf{a}_i^\top \mathbf{x}$ 
3: end for
```

MM Matrix-matrix multiplication

$C = \sum_{i=1}^m \sum_{j=1}^p 2n = 2mnp = \mathcal{O}(mnp)$

Input: $\mathbf{A} \in \mathbb{R}^{m \times n}, \mathbf{B} \in \mathbb{R}^{n \times p}$

Output: $\mathbf{C} = \mathbf{AB}$

```
1: for  $i = 1, \dots, m$  do
2:   for  $j = 1, \dots, p$  do
3:      $c_{ij} = \mathbf{a}_i^\top \mathbf{b}_j$ 
4:   end for
5: end for
```

3 Appendix

Where the error come from calucation:

- If your calculation involves 15 or more significant digits, you will likely encounter rounding errors. (Since $\varepsilon_m \approx 10^{-16}$ (relative accuracy), which is about 15 significant digits.)